Day 4 Exercises: Problem Solving Sessions 1 and 2

This handout combines both Day 4 problem-solving sessions.

Session 1

This session practices translating real contexts into linear programming components. Focus on variables, constraints, objective functions, and feasibility.

1. Define decision variables for making posters and flyers.
2. Write inequalities: $2x + y \leq 120$, $x + 3y \leq 150$, $x, y \geq 0$.
3. Explain in words what each inequality means.
4. State objective for maximizing $P = 5x + 4y$.
5. List two feasible points and one non-feasible point.
6. For points $(0, 0)$, $(20, 30)$, $(40, 10)$, compute $P = 5x + 4y$.
7. Which tested point from Problem 6 is best?
8. Why must nonnegativity constraints be included?
9. Test whether $(30, 25)$ is feasible under the constraints in Problem 2.
10. Write one realistic assumption that makes this LP linear.

Session 2

This session applies the corner-point method and interprets optimal decisions. Focus on feasibility first, then objective values.

1. Graph: $x + y \leq 12$, $x \leq 8$, $y \leq 7$, $x, y \geq 0$.
2. List all corner points of the region.
3. Compute $z = 5x + 4y$ at those corners.
4. State maximizing corner and objective value.
5. Give one real-world interpretation of your result.
6. Add new constraint $x \leq 6$. Recompute candidate corners that change.
7. Under the new constraint in Problem 6, does the optimum change? Explain.
8. Write one sentence describing what an active (binding) constraint means at your optimal point.
9. Reflection: give one reason a graph-based LP method is less practical for many variables.